

Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1750–2024

Data: Global Carbon Budget 2025, NASA

Audience: Policy Analysts & Climate Negotiators

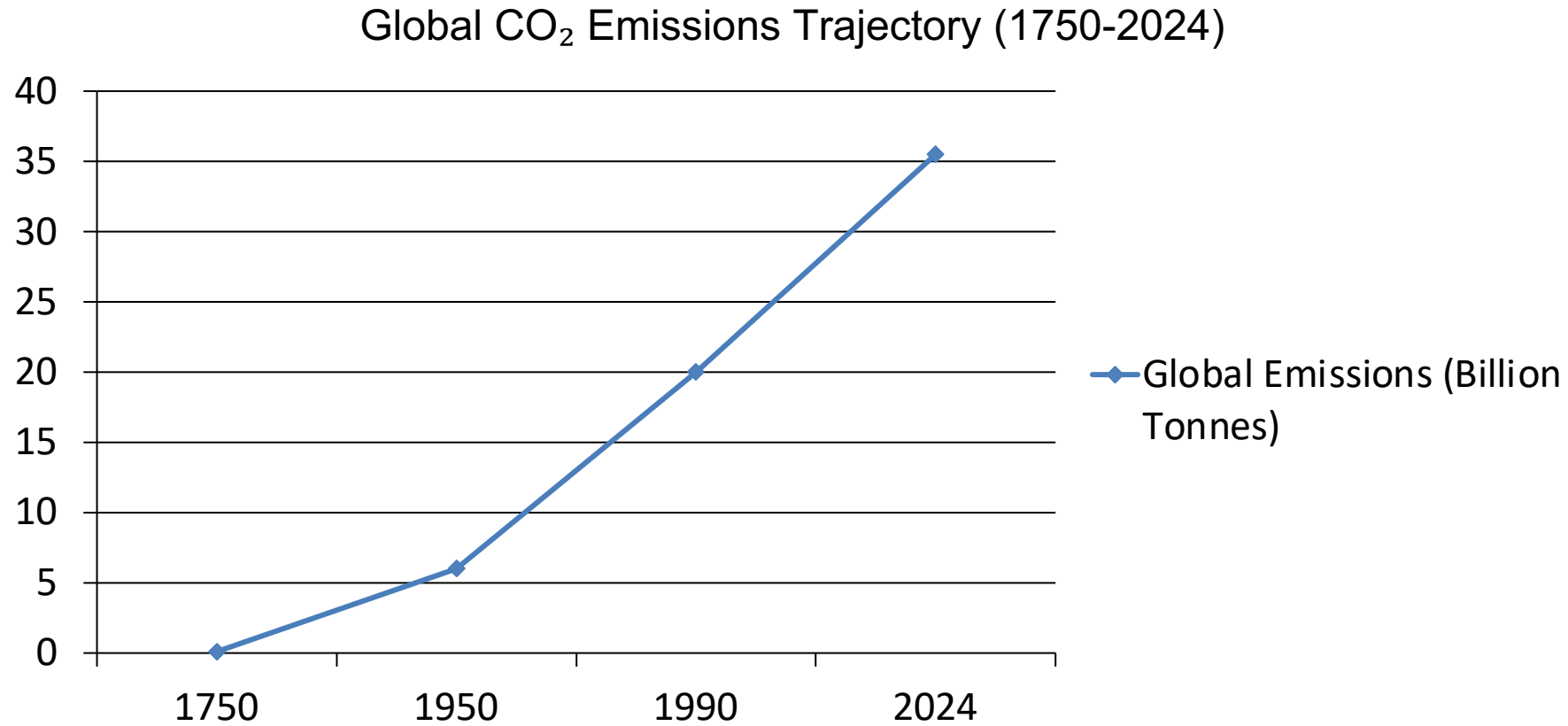
Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

Latest Mauna Loa Measurement (Feb 2026)

- Pre-industrial baseline: ~280 ppm (a 50% increase)
- Current levels are at least 1.5× greater than the natural increase at the end of the last ice age.
- The sharp acceleration since 1958 is linked primarily to burning coal, oil, and natural gas.

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes



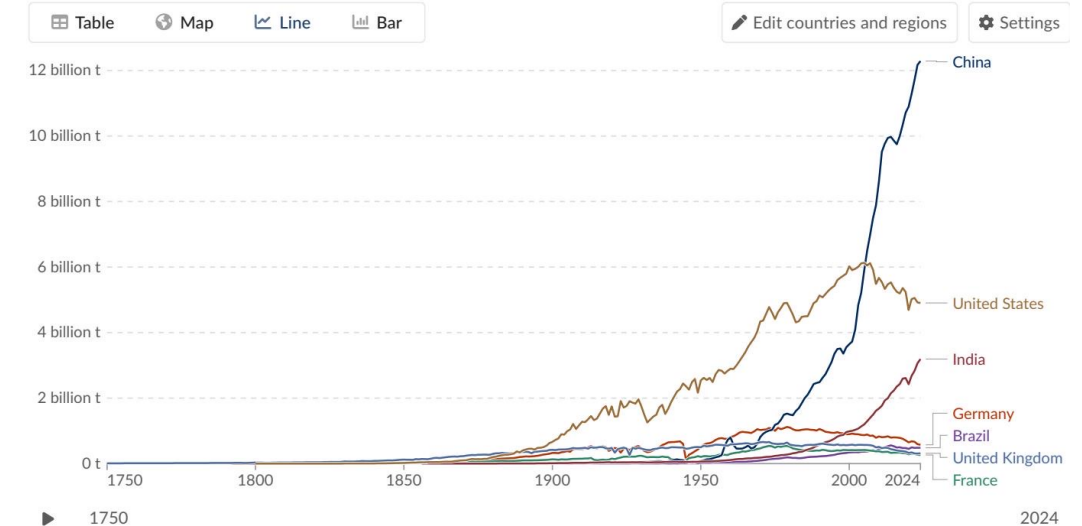
Key Inflection Points: 1950 (6 Bt) → 1990 (20 Bt) → Today (35+ Bt)

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

Country/Region	CO ₂ (Bt)	Share	Trend
China	~12.5	~27%	Slight increase
United States	~5.0	~14%	Declining
India	~3.0	~8%	Rising
EU-27	~2.7	~7%	Declining
Germany	~0.6	~1.7%	Declining
United Kingdom	~0.3	~0.9%	Declining
Brazil	~0.5	~1.4%	Stable

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.



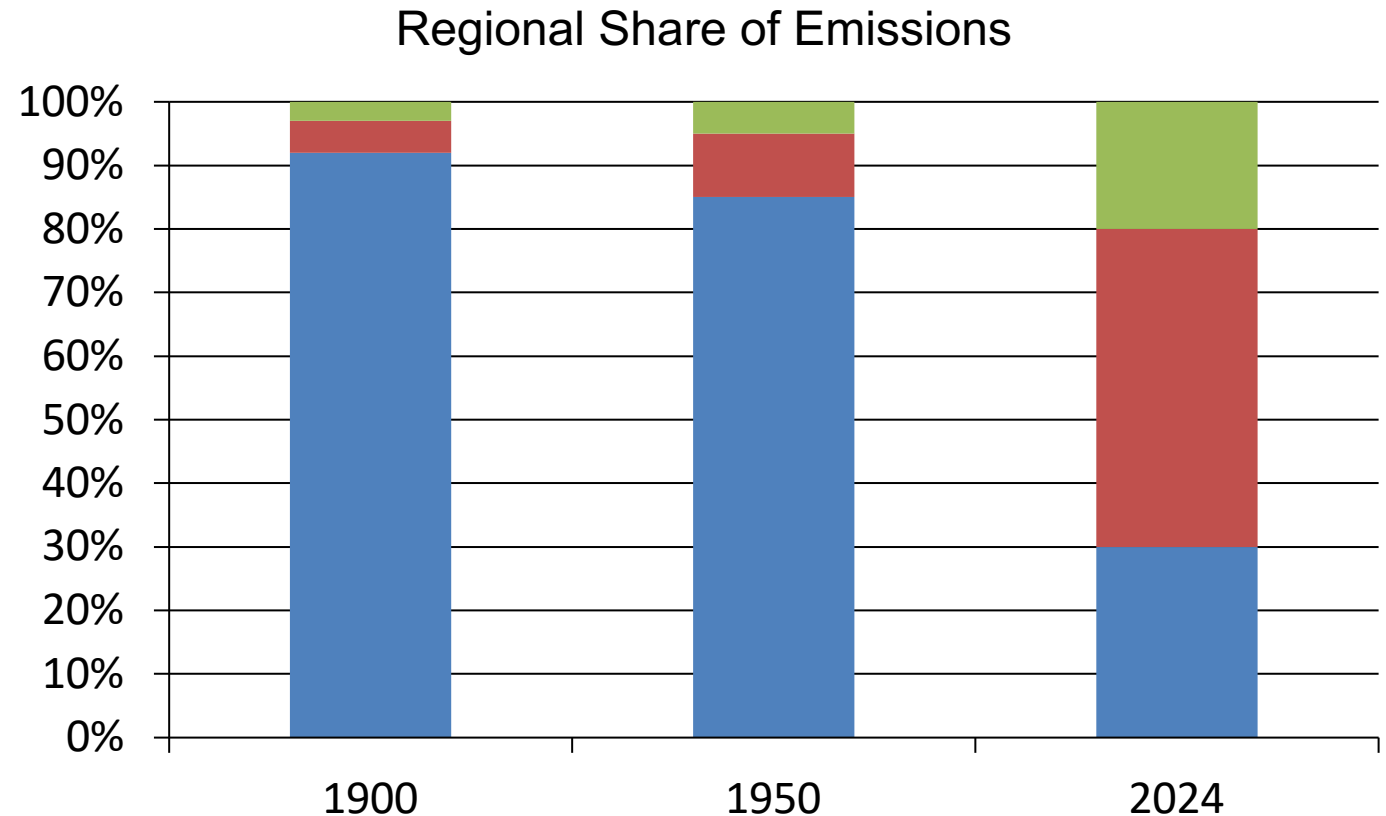
Data source: Global Carbon Budget (2025) – [Learn more about this data](#)
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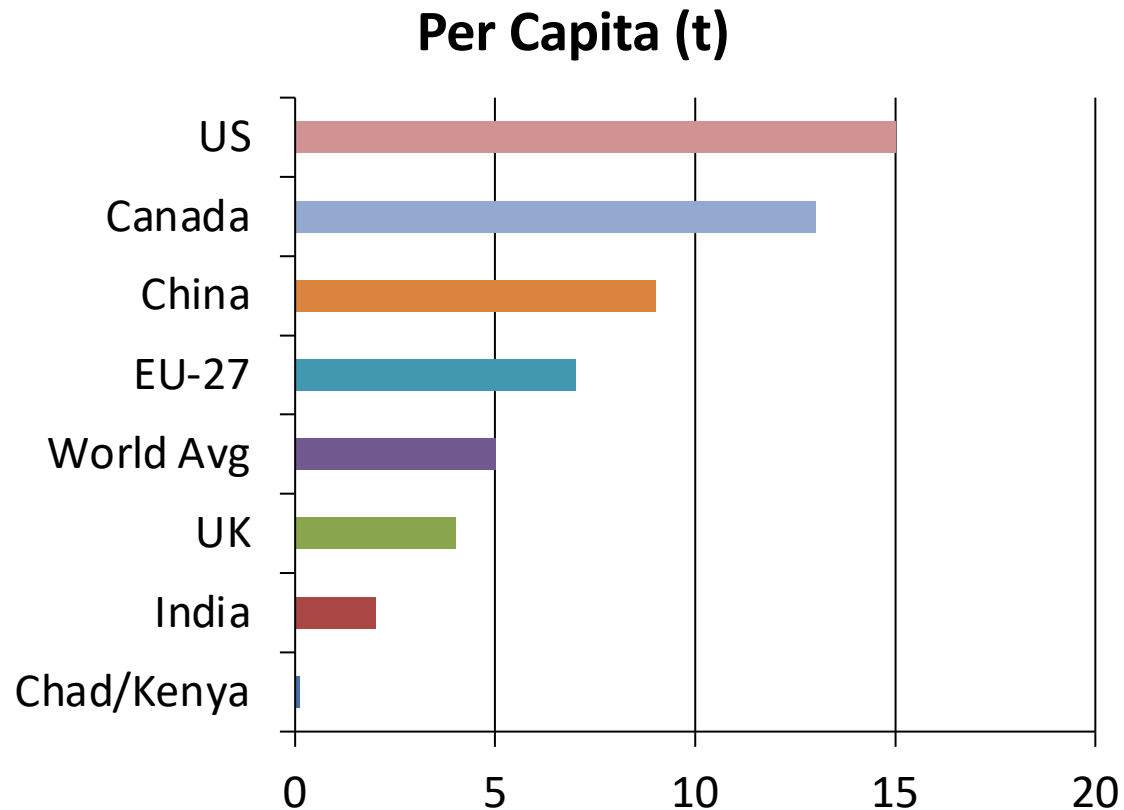
Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

Dramatic Regional Shifts (1900 vs 2024):

- 1900: Europe and the US accounted for > 90% of global emissions.
- 1950: Europe and the US still represented > 85%.
- 2024: Asia (home to ~60% of population) produces ~50% of all emissions.
- 2024: US + Europe combined represent less than one-third of the total.
- Africa and South America each contribute only 3-4% (comparable to international aviation/shipping).



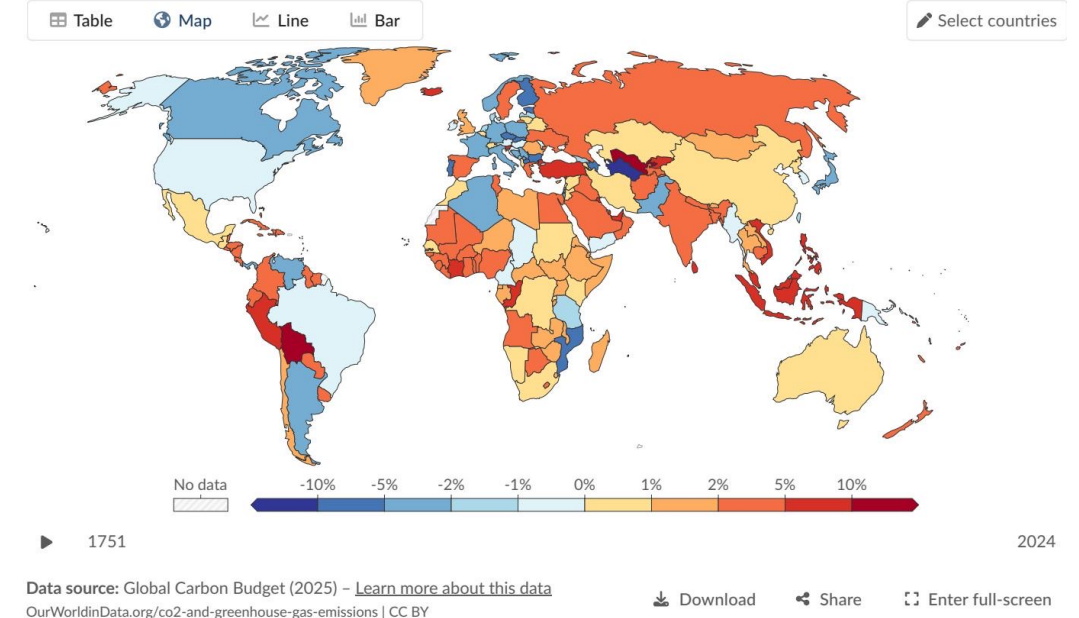
Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



Note: Extreme inequality—an average American emits in under two days what a person in Chad emits in a year.

Annual percentage change in CO₂ emissions, 2024

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

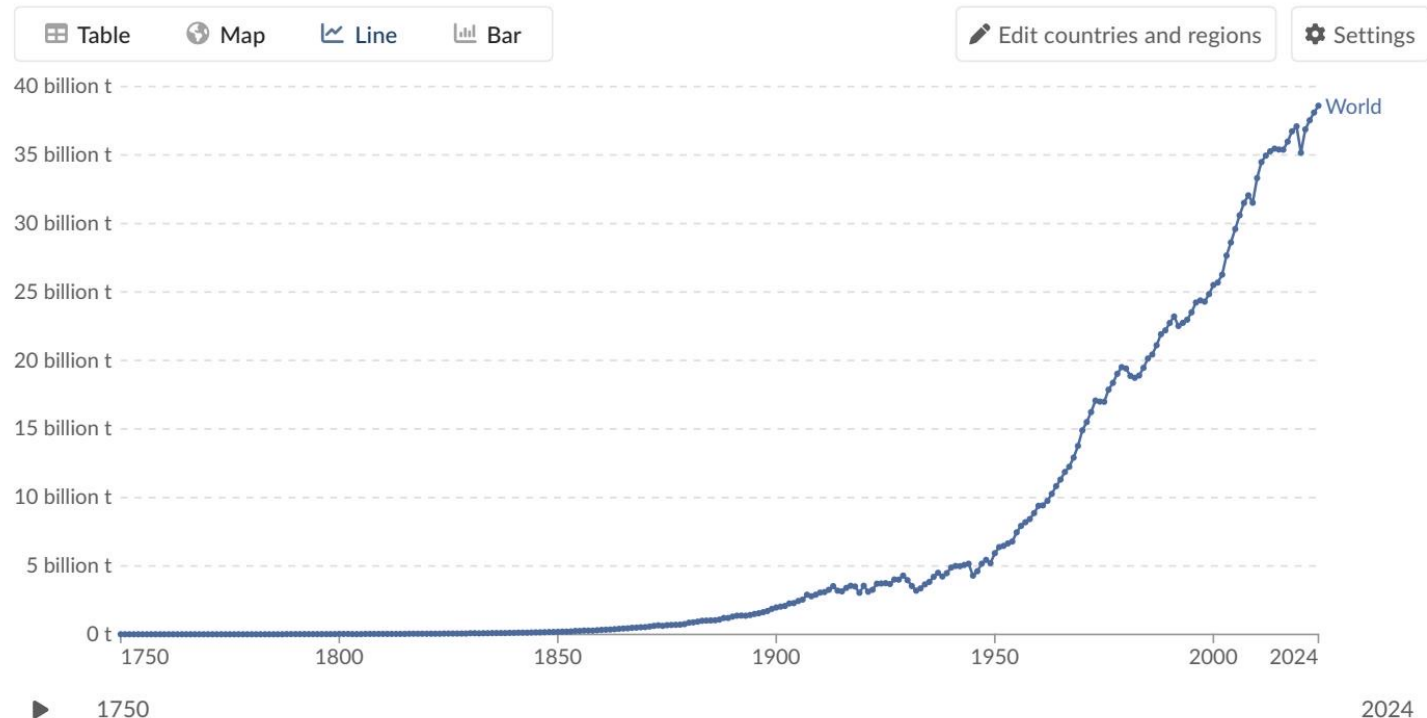


2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

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Regional Patterns in 2024:

- US & Europe (Blue): Declining emissions (-1% to -5%).
- Russia & SE Asia (Orange/Red): Increasing emissions (+2% to +10%).
- Africa & South America: Mixed picture, with some large increases (>10%) and some declines.

Data source: Global Carbon Budget (2025) – [Learn more about this data](#)

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Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Key Insight: Prosperity vs. Emissions

Countries with similar living standards differ significantly based on their energy mix. France and the UK have much lower per capita emissions than Germany or the Netherlands, driven largely by higher shares of nuclear and renewable energy.

Group	Key Countries	Energy Mix & Impact
Lower Emissions	France (~4t), UK (~4t)	High nuclear and renewables
Higher Emissions	Germany (~6t), Netherlands	Higher fossil fuel reliance

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

- Global emissions exceed 35 Bt and have not yet peaked, despite slowing growth.
- China alone emits >25% of the global total (~12.5 Bt).
- Per capita inequalities remain extreme (150× gap between highest and lowest).
- US and Europe are declining, but historical cumulative responsibility remains large.
- Policy and energy mix choices — not just prosperity — determine per capita outcomes.